

Frame transport for framed formal monodromy: the lemma, and why the present receiver cannot host it

Memo for a reader of *Irrationality of Cubic Threefolds after One Stabilization*

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Abstract

Proposition 4.7 of the draft transports framed formal monodromy across the comparison isomorphisms of Iritani and Iritani–Koto. The transport step itself is asserted in one sentence. This memo proves it, in the form it should take, over a differential field; then shows that the coefficient receiver the draft constructs cannot be that field, for a reason that is quantitative rather than cosmetic; and isolates the exact inequality any repair must establish. The inequality holds at the cubic endpoint and is not supplied for a general blow-up center. Section 5 says where each piece belongs in the manuscript.

1 The assertion at issue

Everything below refers to Section 4 of the draft, *Framed formal monodromy*. The relevant statements are Definition 4.1 (the multiplicity ν_6 , p. 15), Definition 4.2 (the pro-Laurent gauge group, p. 15), Lemma 4.3 (coefficientwise base change, p. 16), Lemma 4.5 (formal base shift, p. 16) and Proposition 4.7 (framed operation formulas, p. 18).

Lemma 4.3 ends with the sentence

If the gauge uses only integral powers of z at every finite level, this conjugacy preserves the original- z -disc frame,

and its proof discharges that clause in one sentence: “the deck action fixes every integral power of z at each finite level, so the compatible gauge is single-valued for the framed turn.” Lemma 4.5 then *defines*

$$M^{\text{bulk}} = G M_{\text{f},V}^{\text{small,shifted}} G^{-1} \tag{4.1c}$$

and Proposition 4.7 concludes from it that “each center target summand has the framed monodromy of its specialized small connection.” That conclusion is the content of the proposition, and it does not follow from the displayed definition alone: conjugating a matrix by an arbitrary invertible matrix preserves its characteristic polynomial for trivial reasons, but says nothing about whether the result is the framed operator of the other module. What is missing is a statement that a gauge fixed by the turn transports framed operators, together with an algebra in which the turn acts on solutions. The draft constructs a common coefficient receiver, and the turn acts trivially on it, so the content has to live on the solution side.

2 The lemma, over a differential field

This section is self-contained and, I believe, complete. Notation for Exp_V , Ω_V and the complement $K = \mathbf{C} \oplus V$ is the draft's, from (4.0) and (4.0a).

The ground field

Let Γ be a totally ordered abelian group and let

$$\mathcal{H} = \Omega_V((\Gamma \oplus \mathbf{Z}e_z))$$

be the Hahn field of series $\sum_g c_g x^g$ with well-ordered support and coefficients in Ω_V , the value group ordered lexicographically with the e_z -coordinate *last*. Write $z = x^{e_z}$, and let θ be the derivation multiplying the coefficient at (γ, k) by k ; Leibniz holds because k is additive on the value group. Then $\theta z = z$ and

$$\mathcal{C} := \mathcal{H}^{\theta=0} = \Omega_V((\Gamma)),$$

a field. Taking $\Gamma = 0$ gives $\mathcal{H} = \Omega_V((z))$, the classical case. For $e \geq 1$ put $\mathcal{H}_e = \Omega_V((\Gamma \oplus \frac{1}{e}\mathbf{Z}e_z))$, a degree- e extension of $\mathcal{H}$ containing $w = x^{e_z/e}$ with $w^e = z$, and $\theta_w = e\theta$, so $\theta_w w = w$.

The solution algebra and the turn

Definition 1. The *universal formal solution algebra* $\mathcal{U}_e$ is the $\mathcal{H}_e$ -algebra generated by symbols

$$w^\rho \ (\rho \in \Omega_V), \quad \ell, \quad E(\varphi) \ (\varphi \in w^{-1}\Omega_V[w^{-1}]),$$

subject to $w^\rho w^{\rho'} = w^{\rho+\rho'}$, w^n = the element w^n of $\mathcal{H}_e$ for $n \in \mathbf{Z}$, $E(\varphi)E(\varphi') = E(\varphi + \varphi')$ and $E(0) = 1$, with the derivation extending θ_w by

$$\theta_w(w^\rho) = \rho w^\rho, \quad \theta_w(\ell) = 1, \quad \theta_w(E(\varphi)) = \theta_w(\varphi)E(\varphi).$$

Lemma 2 (Structure). *Fix representatives $\Lambda \subset \Omega_V$ for $\Omega_V/\mathbf{Z}$ with $0 \in \Lambda$. Then $\mathcal{U}_e$ is free as an $\mathcal{H}_e$ -module with basis $\{w^\rho \ell^m E(\varphi) : \rho \in \Lambda, m \geq 0, \varphi\}$. In particular $\mathcal{U}_e \neq 0$ and $\mathcal{H} \subset \mathcal{U}_e$.*

Proof. The w^ρ generate the group algebra of $\Omega_V/\mathbf{Z}$ over $\mathcal{H}_e$, the $E(\varphi)$ that of the additive group $w^{-1}\Omega_V[w^{-1}]$; both groups are torsion-free, so distinct monomials are independent, and ℓ is a polynomial variable. $\square$

Lemma 3 (The turn). *There is a unique automorphism σ of $\mathcal{U}_e$ with*

$$\sigma(w) = e^{2\pi i/e} w, \quad \sigma(w^\rho) = \text{Exp}_V(\rho/e) w^\rho, \quad \sigma(\ell) = \ell + \frac{2\pi i}{e}, \quad \sigma(E(\varphi)) = E(\sigma\varphi),$$

acting on $\mathcal{H}_e$ by multiplying the coefficient at $(\gamma, j/e)$ by $e^{2\pi i j/e}$. It fixes $\mathcal{H}$, hence $\mathcal{C}$, pointwise; it commutes with θ_w ; and on a regular-singular block with residue $R = R_s + R_n$, whose solution is $w^{R_s} e^{R_n \ell}$, the automorphism σ^e acts by $\text{Exp}_V(R_s) \exp(2\pi i R_n) = M_{\text{RS},V}$, the operator of the draft's construction.

Proof. The action on $\mathcal{H}_e$ multiplies coefficients by a character of the value group, hence is a ring automorphism, and it is the identity on $\mathcal{H}$ because there $j \in e\mathbf{Z}$. The two prescriptions agree on integral exponents: $\text{Exp}_V(n/e) = e^{2\pi i n/e}$ by (4.0a), since $n/e \in \mathbf{Q}$. Both σ and θ_w are diagonal in the basis of Lemma 2, and θ_w commutes with $\varphi \mapsto \sigma\varphi$, so they commute. The last claim is the computation $\sigma^e(w^{R_s}) = \text{Exp}_V(R_s) w^{R_s}$ and $\sigma^e(e^{R_n \ell}) = e^{R_n \ell} \exp(2\pi i R_n)$. $\square$

Lemma 4 (Constants). $\mathcal{U}_e^{\theta_w=0} = \mathcal{C}$.

Proof. Let v denote the Hahn valuation of $\mathcal{H}_e$, with values in the lexicographically ordered group $\Gamma \oplus \frac{1}{e}\mathbf{Z}e_z$; it is additive, and θ_w does not enlarge supports, so $v(\theta_w c) \geq v(c)$.

Write $x = \sum c_{\rho,m,\varphi} w^\rho \ell^m E(\varphi)$ in the basis of Lemma 2 and suppose $\theta_w x = 0$. Since $\theta_w(E(\varphi)) = \theta_w(\varphi)E(\varphi)$, the equation splits over φ .

Fix $\varphi \neq 0$ and let M be the top ℓ -degree in that block. For each ρ ,

$$\theta_w(c_{\rho,M}) + \rho c_{\rho,M} + \theta_w(\varphi)c_{\rho,M} = 0.$$

Now $\theta_w(\varphi) \neq 0$ and every monomial of $\theta_w(\varphi)$ has Γ -part 0 and strictly negative e_z -part, so $v(\theta_w(\varphi)c) < v(c) \leq v(\theta_w(c) + \rho c)$ for $c \neq 0$; the lowest term cannot cancel, so $c_{\rho,M} = 0$. Descending in M kills the block.

For $\varphi = 0$ and $\rho \in \Lambda$, the coefficient equations are $\theta_w(c_{\rho,m}) + \rho c_{\rho,m} + (m+1)c_{\rho,m+1} = 0$. On monomials θ_w acts by the integer j attached to the e_z -exponent j/e , so $\theta_w(c) + \rho c = 0$ forces $c = 0$ unless $\rho \in \mathbf{Z}$, that is unless $\rho = 0$. For $\rho = 0$: the top-degree coefficient $c_{0,M}$ is θ_w -constant, and the next equation gives $M c_{0,M} = -\theta_w(c_{0,M-1})$, whose right side has zero component in e_z -degree 0; comparing that component gives $c_{0,M} = 0$ for $M \geq 1$. So only $\rho = 0$, $m = 0$, $\varphi = 0$ survives, with θ_w -constant coefficient, and the θ_w -constants of $\mathcal{H}_e$ are those supported in e_z -degree zero, namely $\mathcal{C}$. $\square$

Framed operators and transport

Let D be a differential module over $\mathcal{H}$, free of rank n , with connection matrix Ω , so that $\nabla = \theta + \Omega$. A *fundamental solution matrix* is $Y \in \mathrm{GL}_n(\mathcal{U}_e)$ with $\theta Y = -\Omega Y$; Levelt–Turrittin supplies one, after a suitable e , for every module over $\Omega_V((z))$, and this is the algebra the draft’s phrase “in the universal formal solution algebra” refers to.

Proposition 5 (The framed operator is the matrix of the turn). *Let Y be a fundamental solution matrix for D . Then*

$$\mathrm{Sol}(D) = \{v \in \mathcal{U}_e^n : \theta v + \Omega v = 0\} = Y\mathcal{C}^n,$$

free of rank n over $\mathcal{C}$ and stable under σ . The matrix M defined by $\sigma(Y) = YM$ lies in $\mathrm{GL}_n(\mathcal{C})$, and replacing Y by another fundamental solution matrix replaces M by a $\mathrm{GL}_n(\mathcal{C})$ -conjugate. For D over $\Omega_V((z))$ in Levelt–Turrittin form, M is the framed operator $M_{\mathbf{f},V}(D)$ of the draft’s construction.

Proof. From $\theta(Y^{-1}) = Y^{-1}\Omega$ one gets $\theta(Y^{-1}v) = Y^{-1}(\theta v + \Omega v)$, so v is horizontal exactly when $Y^{-1}v$ has θ_w -constant entries, and Lemma 4 identifies those with $\mathcal{C}$. Since σ commutes with θ and fixes Ω , whose entries lie in $\mathcal{H}$, it preserves $\mathrm{Sol}(D)$; hence $M = Y^{-1}\sigma(Y)$ has entries in $\mathcal{C}$, and it is invertible with inverse $Y^{-1}\sigma(Y^{-1})^{-1}$. Two fundamental solution matrices differ by right multiplication by an element of $\mathrm{GL}_n(\mathcal{C})$, which conjugates M . The last sentence is the definition: the draft’s operator is by construction the action of the lifted turn on the Levelt–Turrittin solution basis. $\square$

Theorem 6 (Frame transport). *Let D, D' be differential modules over $\mathcal{H}$, free of rank n , with connection matrices Ω, Ω' and fundamental solution matrices $Y, Y' \in \mathrm{GL}_n(\mathcal{U}_e)$. Suppose $G \in \mathrm{GL}_n(\mathcal{H})$ carries D to D' :*

$$\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}.$$

Then GY is a fundamental solution matrix for D' and $\sigma(GY) = (GY)M$, where $\sigma(Y) = YM$. Consequently

$$M_{f,V}(D') = C M_{f,V}(D) C^{-1} \quad \text{for some } C \in \mathrm{GL}_n(\mathcal{C}),$$

so the two framed operators have the same characteristic polynomial in $\mathcal{C}[X]$ and the same multiplicity at every root of unity. Moreover $GM_{f,V}(D)G^{-1}$ is the matrix of the framed turn of D' in the transported frame GY .

Proof. $\theta(GY) = \theta(G)Y - G\Omega Y = -\Omega'(GY)$, using the displayed relation. Since σ fixes $\mathcal{H}$ pointwise, $\sigma(G) = G$, so $\sigma(GY) = G\sigma(Y) = GYM$. Comparing GY with Y' through Proposition 5 gives $Y' = (GY)C_0$ with $C_0 \in \mathrm{GL}_n(\mathcal{C})$, whence $M' = C_0^{-1}MC_0$. The final claim is the identity $\sigma(GY) = (GY)M$ read in the frame GY , whose matrix in the original frame of D' is GMG^{-1} . $\square$

Remark 7 (The hypothesis is exactly “no root of z , no symbol”). Because σ fixes $\mathcal{H}$ pointwise, the invariance $\sigma(G) = G$ is automatic once G has entries in $\mathcal{H}$; the whole content of the hypothesis is that G lies in $\mathcal{H}$ rather than in $\mathcal{U}_e$. No hypothesis-free statement is available: Y itself is a gauge over $\mathcal{U}_e$ carrying D to the trivial module, whose framed operator is the identity. The rank-one case shows what survives. If $g \in \mathcal{H}^\times$ and $\theta(g)/g = \delta \in \Omega_V$, comparing coefficients gives $(k - \delta)g_{(\gamma,k)} = 0$ for every monomial, so $\delta \in \mathbf{Z}$ and g is a constant times z^δ . A gauge over $\mathcal{H}$ shifts residues only by integers, and Exp_V kills those. This is the precise sense in which “integral powers of z ” is the right hypothesis, and it is what the draft’s one-sentence proof was gesturing at.

So the lemma the referee asked for is true and short. The difficulty is elsewhere.

3 The receiver cannot be that field

Theorem 6 needs one ring containing both the comparison gauge and the two modules’ formal solutions. The draft’s receiver, built in the proof of Proposition 4.7, is

$$\mathcal{R}_j = \Omega_{V,j} \otimes_{H_{0,j}} H_{\mathrm{tot},j}, \quad H_{\mathrm{tot},j} = \mathbf{C}((\Gamma_j^{\mathrm{coeff}} \oplus \mathbf{Z}e_z)). \quad (4.4c)$$

It is not such a ring, for three independent reasons.

Finite tensors. An element of $\mathcal{R}_j$ is a *finite* sum $\sum b_\lambda \otimes h_\lambda$. A Levelt–Turrittin gauge is a series $\sum_j P_j w^j$ whose coefficients P_j span an infinite-dimensional $H_{0,j}$ -subspace of $\Omega_{V,j}$ in general. Such a series is not in $\mathcal{R}_j$, and adjoining formal symbols to $\mathcal{R}_j$ does not add it.

The obvious enlargement is not well defined. One would like to map $\mathcal{R}_j$ into the Hahn field $\Omega_{V,j}((\Gamma_j^{\mathrm{coeff}} \oplus \mathbf{Z}e_z))$ by $a \otimes h \mapsto a \cdot h$. This is not well defined. The tensor product over $H_{0,j}$ exists precisely to identify two copies of $H_{0,j}$: one sitting inside $\Omega_{V,j}$ as coefficients, one inside $H_{\mathrm{tot},j}$ as monomial series. In the Hahn field those two copies are different elements, so the map does not respect the relation $ah_0 \otimes h = a \otimes h_0h$. The tensor product is doing real work and cannot be dissolved by hand.

No single Hahn field holds both families. Order the value group with the coefficient degree dominant, as the draft does. Then a pro-Laurent element is legitimate: its z -order runs to $-\infty$ while its coefficient degree grows, so its support is well ordered. But a formal solution whose coefficient degrees run to $-\infty$ at fixed z -power is not. Order it with e_z dominant and the two verdicts swap. The two objects are well ordered for opposite reasons.

The exact criterion

A mixed order is the remaining possibility, and it turns the obstruction into one inequality. Weight a monomial of coefficient degree d and z -power k by $a d + b k$ with $a, b > 0$.

- *Pro-Laurent gauges gain one filtration unit per unit of z -pole.* The recursion (4.1a) in the proof of Lemma 4.5 gives the bulk-degree- m coefficient z -order at least $-m$ and places it in F^m . Its weight grows like $m(a - b)$, so these are well ordered exactly when $a > b$.
- *Splitting gauges lose $w(\Delta\lambda)$ per unit of z -power.* The formal decomposition divides once per order by differences of eigenvalues of the leading matrix $c_1\star$: writing $P = \sum_k P_k z^k$, the splitting equation at order k is $[A_0, P_k] = R_k$ with R_k built from earlier data. Hence P_k has weight at least $-kC$ with $C = w(\Delta\lambda)$, and the weights grow like $k(b - aC)$, well ordered exactly when $b > aC$. Inside a block the regular-singular normalization divides only by residue differences offset by integers, which are nonzero constants and cost nothing.

Both hold together iff $C < 1$, measured in units of the minimal weight ε of a generator of J_j ; that is, iff

$$w(\Delta\lambda) < \varepsilon$$

The ratio is scale-invariant: rescaling the weight function w moves the pro-Laurent gain and the splitting loss by the same factor, so the choice of the integer L in the draft's separation argument is not a free parameter here.

What the criterion depends on

The eigenvalues of $c_1\star$ are homogeneous of quantum degree one, where Q^d has degree $c_1 \cdot d$. When the relevant eigenvalue is a root of a monomial, $\lambda^{c_1 \cdot d_0} \sim Q^{d_0}$, the criterion reads

$$w(Q^{d_0}) < \varepsilon \cdot (c_1 \cdot d_0).$$

Since Q^{d_0} is itself among the generators of J_j , one has $w(Q^{d_0}) \geq \varepsilon$, so the criterion is a Fano-index condition:

- **It holds at the cubic endpoint.** There $c_1 = 2H$ and the exponential factors are governed by $\lambda \sim q^{1/2}$, a square root of a Novikov generator, so $w(\Delta\lambda) = \varepsilon/2$. This is consistent with the draft's own finding that the two rank-one exponential blocks of the cubic are unramified in z , the square root being a coefficient extension.
- **It fails as soon as $c_1 \cdot d_0 = 1$,** since then the criterion demands $w(Q^{d_0}) < \varepsilon$. Proposition 4.7 quantifies over arbitrary smooth centers arising in a weak factorization, and nothing in that hypothesis excludes index one.

- **A center whose small connection is regular singular costs nothing**, $C = 0$: no splitting divisions occur.

So the gap is confined to centers whose small even connection has at least two distinct exponential factors separated by an eigenvalue difference of low fractional index. That is smaller than “all centers”, and it is not empty.

The levelwise route fails for the same reason

It is natural to retreat to finite level, where $(B_j/F^N B_j)((z))$ is an honest Laurent series ring hosting both a bounded-order gauge and a solution algebra with no support conditions at all, and then to pass to the limit of characteristic polynomials rather than of solution algebras. This does not work. The splitting needs $\Delta\lambda$ invertible, and $\Delta\lambda$ lies in J_j , hence is nilpotent at every finite level. It would also place framed monodromy over the nonfield quotients, which the draft deliberately removed.

4 Three routes, and what each must prove

1. **Mixed-order receiver plus a criterion.** Replace (4.4c) by the mixed-order Hahn receiver and prove $w(\Delta\lambda) < \varepsilon$ for the centers the theorem actually uses. The most promising form of this is to show that the centers surviving the draft’s own low-dimensional analysis contribute either no distinct exponential factors or a high enough index.
2. **A solution-free transport theorem.** Prove that the Levelt–Turrittin data of a module over a field — exponential factors, residues modulo $\mathbf{Z}$, deck permutation — are determined by the module after base change along a z -integral isomorphism over $\mathcal{R}_j$, by a slope and valuation argument rather than by comparing solutions. The obstacle to check first: $\mathcal{R}_j$ is not a domain, being a tensor product of two fields, so the e_z -component of the Hahn valuation, which is additive on $H_{\text{tot},j}$, must be shown to behave on the tensor product.
3. **Restrict the proposition.** State Proposition 4.7 for centers satisfying the criterion, and check that the weak factorization used in the main theorem can be run inside that class. This changes the shape of the result and is the author’s call, not a referee’s.

5 Placement in the manuscript

If the mathematics is settled, the edits are localized.

1. **The ground field and solution algebra** go immediately before the sentence “Apply Levelt–Turrittin after a ramification $z = w^e$ ” on p. 14. That sentence already speaks of “the universal formal solution algebra” without having constructed one, so this repairs a forward reference as well.
2. **Lemma 4 and Proposition 5** go directly after Definition 4.1, where the framed operator has just been defined. They convert that definition from a recipe into a characterization: $M_{f,V}$ is the matrix of the turn on the horizontal module.
3. **Theorem 6** becomes a new lemma, placed after the paragraph beginning “The frame matters” on p. 15 and before Definition 4.2. It must carry one semantic label, `lem:frame-transport`.

4. **Lemma 4.3** loses its final asserted sentence and the corresponding sentence of its proof; both become a reference to the new lemma.
5. **Lemma 4.5**, at (4.1c), gains one sentence: by the new lemma, M^{bulk} is the framed operator of the gauged module in the transported frame, not merely a conjugate of the small one.
6. **Proposition 4.7** cites the new lemma at its three load-bearing steps — the appeal to Theorem 5.18 of Iritani’s blow-up paper, the center summand conclusion, and the projective-bundle conclusion — and must say explicitly which field plays the role of $\mathcal{H}$ at each, together with the criterion verified there.

Two mechanical consequences: a new theorem-like environment must carry exactly one semantic label and a matching row in the claim map consumed by the manuscript’s source-only gate, and the coverage snapshot recorded in the two verification READMEs changes with it.

6 Questions I would like answered

1. Is there a receiver that holds both families that we have missed — in particular, a completion of $\mathcal{R}_j$ in which the convolution defining a product of a pro-Laurent gauge with a formal solution converges for structural reasons rather than by a rate comparison?
2. Is route 2 available? A transport theorem that never compares solutions would make the receiver question moot.
3. Failing both: is $w(\Delta\lambda) < \varepsilon$ automatic for the centers that actually arise in a weak factorization between $X \times \mathbf{P}^1$ and a rational model, given the draft’s own vanishing result for centers of dimension at most two?
4. Is the criterion the right invariant at all, or is it an artifact of proving transport through an ordered receiver?